

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

ORDER NO. 94-353

WASTE DISCHARGE REQUIREMENTS

FOR
MADERA COUNTY MAINTENANCE DISTRICT NO. 8A
NORTH FORK WASTEWATER TREATMENT FACILITY
MADERA COUNTY

The California Regional Water Quality Control Board, Central Valley Region, (hereafter Board) finds that:

1. Madera County Engineering Department, on behalf of Madera County Maintenance District No. 8A (hereafter Discharger), submitted a Report of Waste Discharge dated 28 January 1994, and a site evaluation report dated 15 October 1990, for an increase in the volume of waste discharged from 38,000 to 60,000 gallons per day (gpd) from the County Maintenance District No. 8A Wastewater Treatment Facility (WWTF). The WWTF serves the unincorporated community of North Fork.
2. Waste Discharge Requirements Order No. 86-131, adopted by the Board on 27 June 1986, prescribes requirements for a discharge of 38,000 gallons per day (gpd) of treated domestic wastewater from an activated sludge sewage treatment package plant to land.
3. Order No. 86-131 is not adequate for the increase in volume of waste discharged nor consistent with current plans and policies of the Board.
4. In 1988, the Discharger abandoned the activated sludge treatment plant and a storage pond because they created water quality problems due to flooding by Willow Creek, inadequate storage capacity, and effluent runoff into surface waters from a spray field site.
5. In 1988, the Discharger completed construction of a package sewage treatment plant comprised of headworks, a diffused air chamber, a final clarifier, and a sludge digester with treatment capacity for an average dry weather flow of 38,000 gpd and a peak wet weather flow of 230,000 gpd. In 1992, it added an identical 38,000 gpd package plant to handle commercial growth within North Fork. The current WWTF includes headworks, the two package plants, chlorination units, an emergency holding tank, a 23-acre-foot effluent storage pond, and a 20-acre spray field (hereafter "disposal area").
6. Sludge from this site, as well as from other County maintenance district WWTFs, has been disposed of by landspreading on the Wyle Ranch without Board requirements or a waiver thereof.
7. The disposal area and storage pond are in Section 19 and the rest of the WWTF in Section 18, T8S, R23E, MDB&M, with surface water drainage to Willow Creek, as shown in Attachment A, which is attached hereto and part of this Order by reference. The site lies

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within the Wishon hydrologic subarea (No. 540.21), as depicted on interagency hydrologic maps prepared by the Department of Water Resources in August 1986.

8. The Board adopted a Water Quality Control Plan, Second Edition, for the San Joaquin River Basin (hereafter Basin Plan) which contains water quality objectives for waters of the Basin. These requirements implement the Basin Plan.
9. Willow Creek is tributary to the San Joaquin River, a water of the United States. Several ephemeral streams tributary to the creek cut across the disposal area.
10. The beneficial uses of this reach of the San Joaquin River are municipal, industrial, and agricultural supply; recreation; esthetic enjoyment; hydropower generation; and preservation and enhancement of fish, wildlife, and other aquatic resources.
11. The beneficial uses of underlying ground water are domestic, industrial, and agricultural supply.
12. The treatment facilities, emergency holding tank, and storage pond are within the 100-year flood plain of Willow Creek, but protected against a 100-year flood. The disposal area has native vegetation, but it is on a hillside which slopes towards the Creek at up to 35 percent. The area has runoff controls and effluent return systems for flows of 38,000 gpd, to prevent effluent from leaving the disposal area and flowing into ephemeral streams tributary to Willow Creek. The Discharger has not provided evidence that has it similar controls and return systems for a flow of 60,000 gpd.
13. The site evaluation report indicates that the upper two to four feet of soils in the disposal area are silty sands, which are underlain by decomposed granite. Staff review of data in the report indicates that on-site percolation rates range from 2.5 to > 600 min/inch, with an average of about 150 min/inch. The Discharger has not submitted evidence that it has disposal capacity of 60,000 gpd.
14. The Discharger reports there is no irrigated land near the WWTF and the low volume of wastewater discharged makes other forms or reclamation economically infeasible.
15. On-site ground water occurs in fractured granitic zones at depths of up to 900 feet below ground surface (bgs). First encountered ground water is about 250 feet bgs, but springs are not unusual during the rain season. Areal ground water is of good to excellent quality.
16. On 16 August 1980, Madera County certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. Compliance with this Order will prevent any adverse impacts on water quality.

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17. The permitted discharge is consistent with the antidegradation provisions of State Water Resources Control Board Resolution 68-16. This Order provides for an increase in the volume of pollutants discharged. The project's discharge to the disposal area will not cause significant impacts on ground water and surface waters, or depletion of limited ground water resources. The increase in the discharge allows wastewater utility service necessary for the protection of the public and economic expansion in the area, and is considered to be a benefit to the people of the State.
18. The Board has notified the Discharger and interested agencies and persons of its intent to prescribe waste discharge requirements for this discharge and has provided them with an opportunity for a public hearing and an opportunity to submit their written views and recommendations.
19. The Board, in a public meeting, heard and considered all comments pertaining to the discharge.

IT IS HEREBY ORDERED that Order No. 86-131 is rescinded and Madera County Maintenance District No. 8A, its agents, successors, and assigns, in order to meet the provisions contained in Division 7 of the California Water Code and regulations adopted thereunder, shall comply with the following:

A. Discharge Prohibitions:

1. Discharge of wastes to surface waters or surface water drainage courses is prohibited.
2. Bypass or overflow of untreated or partially treated waste is prohibited.
3. Discharge of waste classified as 'hazardous' or 'designated', as defined in Sections 2521(a) and 2522(a) of Title 23, California Code of Regulations, Section 2510, et seq., (hereafter Chapter 15) is prohibited.

B. Discharge Specifications:

1. Until Provision No. E.3 is satisfied, the monthly average dry weather discharge shall not exceed 38,000 gpd.
2. After Provision No. E.3 is satisfied, the monthly average dry weather discharge shall not exceed 60,000 gpd.
3. Objectionable odors originating at this facility shall not be perceivable beyond the limits of the wastewater treatment and disposal areas.

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4. As a means of discerning compliance with Discharge Specification No. B.3, the dissolved oxygen content in the upper zone (1 foot) of wastewater in ponds shall not be less than 1.0 mg/l.
5. The effluent from the treatment facility shall not exceed the following limits:

<u>Constituent</u>	<u>Units</u>	<u>Monthly Average</u>	<u>Daily Maximum</u>
BOD ₅ ¹	mg/l	30	60
Total Suspended Solids	mg/l	30	60
Total Coliform Organisms	MPN/100 ml	23	240
Settleable Solids	ml/l	0.1	0.2

¹ 5-day, 20° Celsius biochemical oxygen demand.

6. Ponds shall not have a pH less than 6.5 or greater than 9.5
7. The discharge to the disposal area shall be managed to minimize erosion, runoff, movement of aerosols from the disposal area. Specifically:
 - a. All wastewater applied to the disposal area must infiltrate completely within a 24-hour period or controlled and returned to the treatment/storage facilities.
 - b. Application of wastewater shall be at reasonable rates considering native grasses and vegetation, soil, slope, climate, and irrigation management system.
 - c. There shall be no discharge to the disposal area while precipitation is occurring or is forecast to occur, and for at least 24 hours after cessation of precipitation.
8. Ponds shall be managed to prevent breeding of mosquitos. In particular:
 - a. An erosion control program should assure that small coves and irregularities are not created around the perimeter of the water surface.
 - b. Weeds shall be minimized through control of water depth, harvesting, or herbicides.
 - c. Dead algae, vegetation, and debris shall not accumulate on the water surface.
9. Public contact with wastewater in the treatment and disposal areas shall be precluded through such means as fences, signs, and other acceptable alternatives.

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10. Ponds shall have sufficient capacity to accommodate allowable wastewater flow and design seasonal precipitation and ancillary inflow and infiltration during the nonirrigation season. Design seasonal precipitation shall be based on total annual precipitation using a return period of 100 years, distributed monthly in accordance with historical rainfall patterns. Freeboard shall never be less than two feet (measured vertically).
11. On or about 1 October of each year, available pond storage capacity shall at least equal the volume necessary to comply with Discharge Specification No. B.10.

C. Sludge Disposal Specifications:

1. Collected screenings, sludges, and other solids removed from liquid wastes shall be disposed of in a manner that is consistent with Chapter 15 and approved by the Executive Officer.
2. Any proposed change in sludge use or disposal practice shall be reported to the Executive Officer at least 90 days in advance of the change.
3. Use and disposal of sewage sludge shall comply with existing federal and state laws and regulations, including permitting requirements and technical standards included in 40 CFR 503.

If the State Water Resources Control Board and the regional water quality control boards are given the authority to implement regulations contained in 40 CFR 503, this Order may be reopened to incorporate appropriate time schedules and technical standards. The Discharger must comply with the standards and time schedules contained in 40 CFR 503 whether or not they have been incorporated into this Order.

D. Ground Water Limitations:

The discharge, in combination with other sources, shall not cause underlying ground water to:

1. Contain waste constituents in concentrations statistically greater than receiving water limits specified below.
2. Contain chemicals, heavy metals, or trace elements in concentrations that adversely affect beneficial uses or exceed maximum contaminant levels specified in the California Code of Regulations, Title 22, Division 4, Chapter 15.

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3. Equal or exceed a most probable number of total coliform organisms of 2.2/100 ml over any seven-day period.
4. Contain taste or odor-producing substances in concentrations that cause nuisance or adversely affect beneficial uses.

E. Provisions:

1. The Discharger shall comply with Monitoring and Reporting Program No. 94-353, which is part of this Order, and any revisions thereto as ordered by the Executive Officer.
2. The Discharger shall comply with the "Standard Provisions and Reporting Requirements for Waste Discharge Requirements," dated 1 March 1991, which are attached hereto and by reference a part of this Order. This attachment and its individual paragraphs are commonly referenced as "Standard Provision(s)."
3. The Discharger shall provide a technical report that certifies the WWTF has treatment, storage, and disposal capacity of 60,000 mgd and includes an irrigation management plan for 60,000 gpd to satisfy Discharge Specification B.2. The irrigation management plan shall include a water balance for current and design flows, with storage requirements using total annual precipitation data with a 100-year return frequency for the storage pond, and measures/controls to prevent effluent runoff from leaving the disposal area and entering surface waters and surface drainage courses. The plan shall also demonstrate that disposal can be accomplished at discharge rates commensurate with the infiltration and percolation capacity of on-site soils, and evapotranspiration rates of the native grasses and vegetation. The report shall be subject to approval by the Executive Officer and shall be prepared and signed by a California registered civil engineer experienced in wastewater treatment and disposal. If the report meets the satisfaction of the Executive Officer, he shall notify the Discharger that this Provision is satisfied and Discharge Specification B.2 is in effect.
4. By 15 January 1995, the Discharger shall submit a Report of Waste Discharge and a work plan for sludge disposal. The work plan shall address compliance with Sludge Disposal Specification Nos. C.1 to C.3. The plan shall include a time schedule for implementation and be prepared by a California registered civil engineer experienced in wastewater treatment and disposal.
5. In the event of any change in control or ownership of land or waste discharge facilities described herein, the Discharger shall notify the succeeding owner or operator of the existence of this Order by letter, a copy of which shall be immediately forwarded to this office.

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To assume operation under this Order, the succeeding owner or operator must apply in writing to the Executive Officer requesting transfer of the Order. The request must contain the requesting entity's full legal name, the State of incorporation if a corporation, the name and address and telephone number of the persons responsible for contact with the Board, and a statement. The statement shall comply with the signatory paragraph of Standard Provision B.3 and state that the new owner or operator assumes full responsibility for compliance with this Order. Failure to submit the request shall be considered a discharge without requirements, a violation of the California Water Code. Transfer shall be approved or disapproved by the Executive Officer.

6. At least 90 days prior to termination or expiration of any lease, contract, or agreement involving disposal or reclamation areas or off-site reuse of effluent, used to justify the capacity authorized herein and assure compliance with this Order, the Discharger shall notify the Board in writing of the situation and of what measures have been taken or are being taken to assure full compliance with this Order.
7. The Discharger shall use the best practicable cost-effective control technique currently available to comply with salinity limits specified in this order.
8. The Discharger must comply with all conditions of this Order, including timely submittal of technical and monitoring reports as directed by the Executive Officer. Violations may result in enforcement action, including Regional Board or court orders requiring corrective action or imposing civil monetary liability, or in revision or rescission of this Order.
9. The Discharger shall submit to the Board on or before each report due date the specified document or, if an action is specified, a written report detailing evidence of compliance with the date and task. If noncompliance is being reported, the reasons for such noncompliance shall be stated, plus an estimate of the date when the Discharger will be in compliance. The Discharger shall notify the Board by letter when it returns to compliance with the time schedule.
10. A copy of this Order shall be kept at the WWTF for reference by operating personnel. Key operating personnel shall be familiar with its contents.
11. If reclaimed water is used for construction purposes, it shall comply with the most current edition of "Guidelines for Use of Reclaimed Water for Construction Purposes". Other uses of reclaimed water not specifically authorized herein shall be subject to the approval of the Executive Officer and shall comply with 22 CCR, Division 4.

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12. The Board will review this Order periodically and will revise requirements when necessary.

I, WILLIAM H. CROOKS, Executive Officer, do hereby certify the foregoing is a full, true, and correct copy of an Order adopted by the California Regional Water Quality Control Board, Central Valley Region, on 9 December 1994.



WILLIAM H. CROOKS, Executive Officer

JLA:jla/fmc:12/9/94

CALIFORNIA REGIONAL WATER QUALITY CONTROL BOARD
CENTRAL VALLEY REGION

MONITORING AND REPORTING PROGRAM NO. 94-353
FOR
MADERA COUNTY MAINTENANCE DISTRICT NO. 8A
NORTH FORK WASTEWATER TREATMENT FACILITY
MADERA COUNTY

Specific sample station locations shall be established with concurrence of the Board's staff, and a description of the stations shall be submitted to the Board by 15 January 1995 and attached to this Program.

EFFLUENT MONITORING

Except for flow, which may be measured at the headworks or the treatment unit, effluent samples shall be collected prior to discharge to the storage pond. Effluent samples should be representative of the volume and nature of the discharge. Time of collection of a grab sample shall be recorded. Effluent monitoring shall include at least the following:

<u>Constituents</u>	<u>Units</u>	<u>Type of Sample</u>	<u>Frequency¹</u>
Total Daily Flow	gpd	Continuous	Daily
Settleable Solids	ml/l	Grab	Daily
Specific Conductivity @ 25°C	μmhos/cm	Grab	Weekly
BOD ₅ ²	mg/l	Grab	Weekly
Total Suspended Solids	mg/l	Grab	Weekly
Total Coliform Organisms	MPN/100 ml	Grab	Weekly

¹ If results of monitoring a pollutant appear to violate effluent limitations, but monitoring frequency is not sufficient to validate violation (e.g., the monthly average for BOD), or indicate a violation and potential upset of the treatment process (e.g., less than minimum D.O.), the frequency of sampling shall be increased to confirm the magnitude and duration of violation, if any, and aid in identification and resolution of the problem.

² Five-day, 20° Celsius biochemical oxygen demand.

POND MONITORING

The freeboard shall be monitored on the storage pond to the nearest tenth foot. Pond water monitoring shall include the following :

<u>Constituent</u>	<u>Unit</u>	<u>Measurement</u>	<u>Frequency</u>
Freeboard	feet	Observation	Weekly
pH	pH units	Grab	Daily
Dissolved Oxygen ¹	mg/l	Grab	Daily

- ¹ Samples shall be collected at a depth of one foot from each pond, opposite the inlet, and analyzed for dissolved oxygen. Samples shall be collected between 0800 and 0900 hours.

Permanent markers (e.g., staff gages) shall be placed in the pond with calibration indicating the water level at design capacity and available operational freeboard. In addition, the Discharger shall inspect the conditions of the pond once per week and write visual observation in a bound log book. Notations shall include observations of whether weeds are developing in the water or along the bank, and their locations; whether dead algae, vegetation, scum, or debris are accumulating on the pond surface and their location; whether burrowing animals or insects are present; and the color of the pond (e.g., dark sparkling green, dull green, yellow, grey, tan, brown, etc.) A copy of the entries made in the log during each month shall be submitted along with the monitoring report the following month. Where the O&M manual indicates remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

DISPOSAL AREA MONITORING

The Discharger shall inspect the disposal area weekly from November to April and monthly the rest of the year. In conducting the inspection, a log shall be kept of the condition of the disposal area. Special attention shall be given to the presence or absence of vectors, objectional odors, erosion, and effluent runoff. A summary, addressing the conditions of the disposal area, shall be included with the monthly monitoring report. Where remedial action is necessary, the Discharger shall briefly explain in the transmittal what action has been taken or is scheduled to be taken.

REPORTING

Monthly monitoring containing results of all sampling for the month shall be submitted to the Board by the 20th day of the following month.

In reporting the monitoring data, the Discharger shall arrange the data in tabular form so that the date, the constituents, and the concentrations are readily discernible. The data shall be summarized in such a manner that illustrates clearly whether the Discharger complies with waste discharge requirements, including calculation of all averages, etc.

If the Discharger monitors any pollutant at the locations designated herein more frequently than is required by this Order, the results of such monitoring shall be included in the discharge monitoring report.

The Discharger may also be requested to submit an annual report to the Board with tabular and graphical summaries of the monitoring data obtained during the previous year. Any such request shall be made in writing. The report shall discuss the corrective actions taken and planned to bring the discharge into full compliance with the waste discharge requirements.

By **31 January of each year**, the Discharger shall submit a written report to the Executive Officer containing the following:

- a. The names, titles, certificate grade (if any) and general responsibilities of persons operating and maintaining the wastewater treatment plant.
- b. The names and telephone numbers of persons to contact regarding the plant for emergency and routine situations.
- c. A certified statement of when the flow meter and other monitoring instruments and devices were last calibrated, including identification of who did the calibration (Standard Provision C.4).
- d. A statement whether the current operation and maintenance manual, and contingency plan, reflect the wastewater treatment plant as currently constructed and operated, and the dates when these documents were last reviewed for adequacy.
- e. The total quantity of sludge disposed of during the previous year and ultimate disposal site(s).

All reports submitted in response to this Order shall comply with the signatory requirements in Standard Provision B.3.

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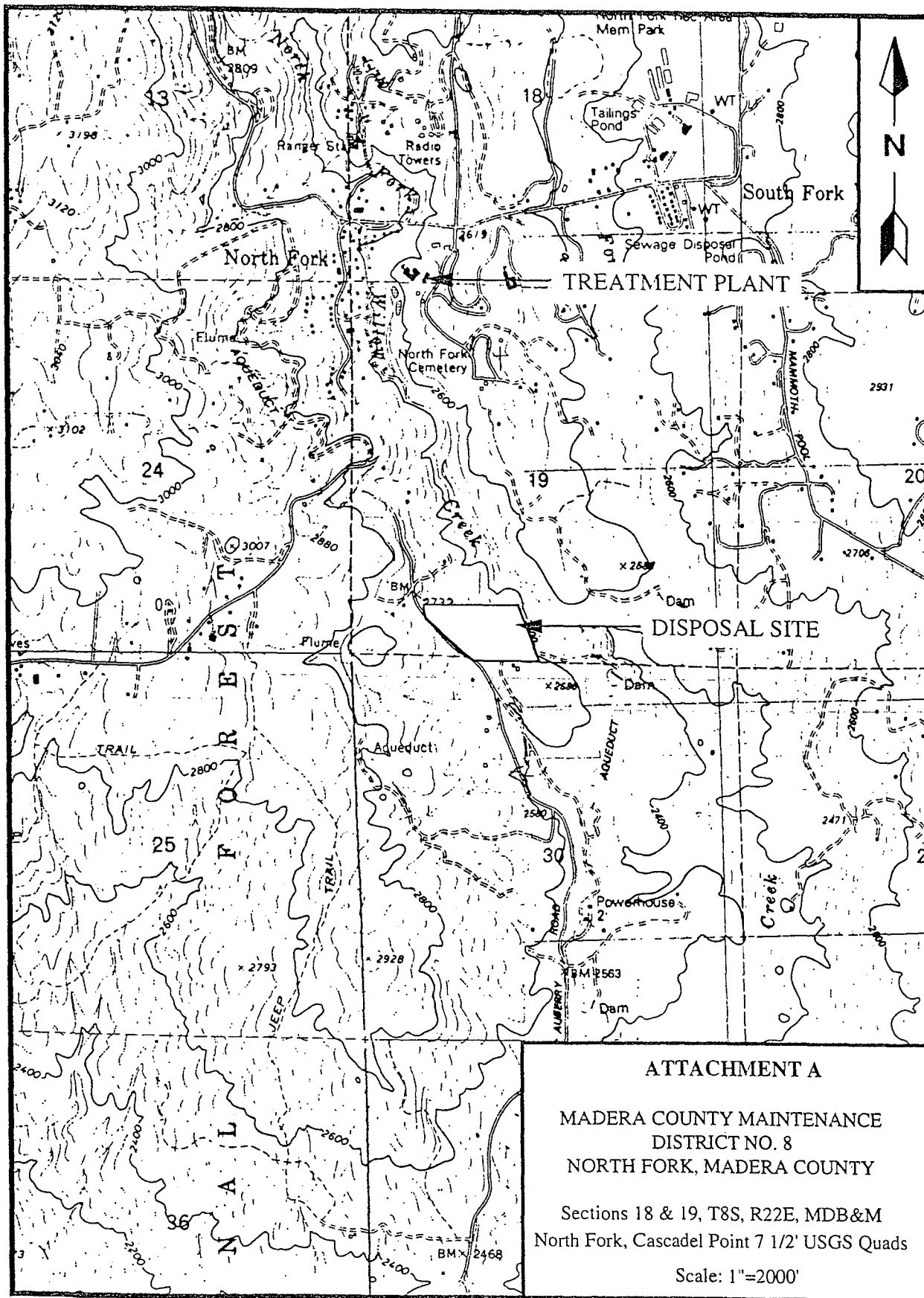
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The Discharger shall implement the above monitoring program on the first day of the month following effective date of this Order.

Ordered by: William H. Crooks
WILLIAM H. CROOKS, Executive Officer

9 December 1994
(Date)

JLA:jla/fmc:12/9/94



INFORMATION SHEET

MADERA COUNTY MAINTENANCE DISTRICT NO. 8A, NORTH FORK WWTF MADERA COUNTY

Madera County Maintenance District No. 8A (hereafter Discharger) applied for revised waste discharge requirements for an increase in the volume of waste discharged from 38,000 to 60,000 gallons per day (gpd) from the Wastewater Treatment Facility (WWTF) serving the unincorporated community of North Fork. Waste Discharge Requirements Order No. 86-131 prescribes secondary and disinfection effluent limits for a discharge of 38,000 gpd of treated domestic wastewater from an activated sludge sewage treatment package plant to land for final disposal by sprinkler irrigation. The existing WWTF includes headworks, two extended aeration sewage treatment package plants, chlorination units, an emergency holding tank, a 23 acre-foot effluent storage pond, and a 20-acre spray field (hereafter "disposal area"). Current flows average 32,000 gpd.

Effluent from the WWTF is pumped about 7000 feet to the storage pond and pumped from there to the hillside spray disposal area. The disposal area has native vegetation, but it is on a hillside which slopes towards Willow Creek at up to 35 percent. Willow Creek is tributary to the San Joaquin River, a water of the United States. Several ephemeral streams tributary to the creek cut across the disposal area. The area has runoff controls and return systems for flows of 38,000 gpd, to prevent effluent from leaving the disposal area and flowing into ephemeral streams tributary to the Creek.

Data submitted by the Discharger indicates that the upper two to four feet of soils in the disposal area are silty sands, which are underlain by decomposed granite. Staff review of the data indicates that on-site percolation rates are 2.5 to > 600 min/inch, with an average of about 150 min/inch. The Discharger has not submitted evidence that it has adequate runoff controls, storage, or disposal capacity of 60,000 gpd. If the Discharger demonstrates that disposal of 60,000 gpd can be properly accomplished at the site, this tentative Order allows for a discharge of 60,000 gpd. Otherwise, the Order limits the flow to 38,000 gpd.

The Discharger does not have an approved sludge disposal plan. Sludge from this site, as well as from other County maintenance district WWTFs, has been disposed of by landspreading on the Wyle Ranch without Board requirements or a waiver thereof. This Order requires the Discharger to apply for requirements for sludge disposal and a sludge management plan for review and approval to dispose of its sludge in accordance with state and federal laws.

Average annual precipitation and evaporation for the area are 35 and 50 inches. On-site ground water occurs in fractured granitic zones at depths of up to 900 feet below ground surface (bgs). First encountered ground water is about 250 feet bgs, but springs are not unusual during the rain season. Areal ground water is of good to excellent quality.

On 16 August 1980, Madera County certified a final environmental impact report (EIR) in accordance with the California Environmental Quality Act (CEQA) (Public Resources Code Section 21000, et seq.) and the State CEQA Guidelines. The project as approved will not have a significant effect on water quality. Compliance with this Order will prevent any adverse impacts on water quality.